Week 7
Packages, String Handling and File I/O
Program 19
Write a Java program to create and use a user-defined package.

Program 1: Save as Message.java

```
package mypackage;

public class Message {
   public void display() {
      System.out.println("Welcome to User-Defined Package");
   }
}
```

Program 2: Save as TestPackage.java

```
import mypackage.Message;

class TestPackage {
   public static void main(String[] args) {
      Message m = new Message();
      m.display();
   }
}
```

OUTPUT:
Welcome to User-Defined Package

Program 20
Write a Java program demonstrating commonly used String methods.

```
class StringDemo {
   public static void main(String[] args) {
      String s = "Hello Java";

      System.out.println("Length: " + s.length());
      System.out.println("Uppercase: " + s.toUpperCase());
      System.out.println("Lowercase: " + s.toLowerCase());
      System.out.println("Character at index 1: " + s.charAt(1));
      System.out.println("Substring: " + s.substring(6));
      System.out.println("Contains Java: " + s.contains("Java"));
      System.out.println("Replace: " + s.replace("Java", "World"));
      System.out.println("Equals: " + s.equals("Hello Java"));
   }
}
```

OUTPUT:
Length: 10
Uppercase: HELLO JAVA
Lowercase: hello java

Character at index 1: e
Substring: Java
Contains Java: true
Replace: Hello World
Equals: true

Program 21

Write a Java program to read data from a file and write data into another file.

```
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;

class FileDemo {
   public static void main(String[] args) {
      try {
         FileReader fr = new FileReader("input.txt");
         FileWriter fw = new FileWriter("output.txt");

         int ch;
         while ((ch = fr.read()) != -1) {
            fw.write(ch);
         }
         fr.close();
         fw.close();

         System.out.println("Data copied successfully.");
      } catch (IOException e) {
         System.out.println("File error: " + e.getMessage());
      }
   }
}
```

Input file (input.txt):
Welcome to Java Programming.

Output on screen:
Data copied successfully.

Output file (output.txt):
Welcome to Java Programming.